

# Nafiz Imtiaz Khan

Davis, California, United States

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## Summary

PhD candidate in Computer Science at UC Davis, advised by Professor Vladimir Filkov. Research focuses on open-source software sustainability, health informatics, and applied data science. Specializes in fine-tuning deep neural networks and large language models, building agentic retrieval-augmented generation (RAG) systems, and developing task-specific AI/ML pipelines.

## Education

### University of California, Davis

PhD in Computer Science | MSc in Computer Science (2025)

Master's degree conferred July 2025, CGPA: 3.91/4.00

September 2023 – Present

California, United States

### Military Institute of Science and Technology (MIST)

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

## Technical Skills

**Programming Languages:** Python, SQL, C/C++, Java, JavaScript, PL/SQL, Typescript, HTML / CSS, Bash

**Frameworks/Libraries :** PyTorch, Keras, Tensorflow, Hugging Face Transformers, LangChain, Scikit-learn, Pandas, NumPy, Matplotlib, BeautifulSoup, Pyvis, Node, React

**ML/AI Techniques:** Transformer architectures, fine-tuning, PEFT, LoRA/QLoRA, multimodal & vision-language models (VLMs), retrieval-augmented generation (RAG), benchmarking & evaluation

**Tools/Platform:** Ollama, Flask, Git, AWS Bedrock, Google Colab, GCP, Docker, Kubernetes, Grafana, Selenium, Tableau, Microsoft Azure, JIRA

**Database:** ChromaDB, FAISS, MongoDB, FireBase, MySQL, Oracle, SQLite, Neo4j

## Professional Experience

### Artificial Intelligence Intern

BetterHelp

June 2026 – Present

San Jose, California, United States (Remote)

- Contributing to AI/ML initiatives as part of the Artificial Intelligence Intern team at BetterHelp.

### Human Frontier Collective Specialist – GenAI (Internship)

Scale AI, Inc.

August 2025 – September 2025

Remote – California, United States

- Designed and critiqued complex, domain-specific problem sets to rigorously test cutting-edge generative AI models, identifying limitations and refining performance.
- Collaborated with Scale researchers in interdisciplinary sessions to analyze model behavior, shape AI research directions, and explore experimental applications.

### Graduate Student Researcher

University of California, Davis

September 2023 – Present

California, United States

- Built **AngioVision** (<https://github.com/Nafiz43/AngioVision>), a multimodal **large language model (LLM)** pipeline for angiography report generation and validation; fine-tuned a **vision-language** framework integrating a **CLIP** alignment module, embedding, and decoder layers, achieving a **10% F1 gain** over the prior SOTA, **Claude 4.6**.
- Led the end-to-end development of **RepoWise** (<https://repowise.github.io/RepoWise-website/>), a domain-aware **retrieval-augmented generation (RAG)** system for OSS governance analytics; benchmarked and demonstrated 84% accuracy on factual queries, outperforming general-purpose LLMs (**GPT-4o**, **Claude 4.5 Sonnet**) on both factual and non-factual repository-analysis tasks.
- Spearheaded the development of **OSSPREY** (<https://oss-prey.github.io/OSSPREY-Website/>), a tool to assess open-source project health via real-time analytics with an average latency of 3.5s and 124MB memory usage per project, integrating **Vue.js** (frontend), **TensorFlow** (deep learning), and **LangChain & Ollama** (LLM inference pipeline design)
- Engineered **ReACT-GPT** (<https://github.com/Nafiz43/ReACT-GPT>), an **LLM-and-RAG**-powered framework for extracting actionable insights from scientific articles, achieving a 75% BERTScore by integrating **LangChain** (orchestration), **ChromaDB** (vector retrieval), and **Ollama** (lightweight LLM inference)
- Built **EvidenceBot** (<https://nafiz43.github.io/EvidenceBot/>), a privacy-preserving **RAG** tool for querying and extracting insights from a diverse set of documents, using **Streamlit** and the **ReACT-GPT** framework
- Created **PCL-Fetcher** (<https://nafiz43.github.io/PCL-Fetcher/>), an LLM-powered pipeline automating procedural case log documentation from radiology reports, achieving an 86% F1 score—exceeding the crosswalk benchmark by over 13%
- Designed **MammoGen-RAG** (<https://github.com/Nafiz43/MammoGen-RAG>), a multimodal RAG pipeline leveraging **vision-language models (VLMs)** to automate mammography report generation, achieving a 90% F1 score through integrated visual and textual reasoning
- Improved foundational language model performance in **LLM-as-Developers** (<https://github.com/zeerakb1/llm-developers-evaluation>) by 10% (BERTScore) on answering StackOverflow questions via **parameter-efficient fine-tuning (PEFT)** and **QLoRA**
- Developed and optimized **Transformer-** and **LSTM-based** models for forecasting Open Source Software (OSS) project sustainability (<https://github.com/OSS-PREY/OSSPREY-Pex-Forecaster>), achieving an 8% improvement in F1 score over SOTA

- Built the **AFMC Admission Test** Module, a full-stack software solution for the AFMC Entrance exam, utilizing **Node.js**, **Bootstrap**, and **MongoDB** to generate unique randomized questions for 30,000 candidates and automate the evaluation and ranking process
- Spearheaded a team to develop **C-Archive** using **PHP**, **MySQL**, **SQL**, and **Bootstrap** to store ongoing events of the department [[GITHUB](#)]

## Patents

1. **Nafiz Imtiaz Khan**, Vladimir Filkov, "Systems And Methods For Artificial Intelligence-Driven Evidence-Based Action" (USPTO Filing ID: 32417/70825P) [[REF](#)]

## Award & Honors

- **Best Paper Award** for "RepoWise: A Conversational Framework for Open Source Repository Analysis with Retrieval-Augmented Generation and Structured Analytics," by Sankalp Kashyap, Arjun Ashok, **Nafiz Imtiaz Khan**, and Vladimir Filkov, at the 2nd International Conference on Software Engineering of Emerging Technologies (**SEET 2026**)
- **Champion** in the **Medical Robotics Challenge for Contagious Diseases**, organized by **Imperial College London** [[REF](#)]
- **Champion** in the Creative App Contest of the **Tri Robo Cup Championship** organized by the **MIST Robotics club** [[REF](#)]
- **3rd runner-up** at **COVID-19 Combatants Unification Competition**, organized by **IEEE Bangladesh Chapter** [[REF](#)]
- **Top Downloaded Article Award** - in the year 2021, and 2022; provided by **Engineering Reports Journal - Wiley**
- **GGCS Summer Ph.D Fellowship** - 2024, awarded by Graduate Group of Computer Science, UC Davis
- Recognized for academic excellence at MIST, achieving placement on the **Dean's List of Honor** and receiving the **Merit Scholarship** in 2018, 2019, and 2020

## Publications

[\[Google Scholar\]](#)

1. **Nafiz Imtiaz Khan**, Priyal Soni, Arjun Ashok, Vladimir Filkov, "OSSPREY: AI-Driven Forecasting and Intervention for OSS Project Sustainability," ASE 2025 Tool Demonstrations (IEEE/ACM International Conference on Automated Software Engineering), 2025. [[LINK](#)]
2. **Nafiz Imtiaz Khan**, Kiley Cleland, Vladimir Filkov, and Roger Eric Goldman, "Intelligent Documentation in Medical Education: Can AI Replace Manual Case Logging?," JAMIA Open, 2026. [[LINK](#)]
3. Arjun Ashok\*, **Nafiz Imtiaz Khan**\*, Swati Singhvi, Stefan Stanciulescu, Zhouhao Wang, Vladimir Filkov (\*equal contribution), "Multi-Ecosystem Modeling of OSS Project Sustainability," Journal of Systems and Software (JSS), 2026. [[LINK](#)]
4. **Nafiz Imtiaz Khan**, Vladimir Filkov, "EvidenceBot: A Privacy-Preserving, Customizable RAG-Based Tool for Enhancing Large Language Model Interactions," FSE 2025 Tool Demonstrations (Foundations of Software Engineering), 2025. [[LINK](#)]
5. **Nafiz Imtiaz Khan**, Vladimir Filkov, Roger Eric Goldman, "AI-Driven Automation of Procedural Case Log Documentation," AMIA 2025 Clinical Informatics Conference (abstract), 2025. [[LINK](#)]
6. **Nafiz Imtiaz Khan**, Vladimir Filkov, "From Models to Practice: Enhancing OSS Project Sustainability with Evidence-Based Advice," FSE 2024 (Foundations of Software Engineering), 2024. [[LINK](#)]
7. **Nafiz Imtiaz Khan**, Vladimir Filkov, "Leveraging Language Models to Discover Evidence-Based Actions for OSS Sustainability," (Under review). [[LINK](#)]
8. Sankalp Kashyap, Arjun Ashok, **Nafiz Imtiaz Khan**, Vladimir Filkov, "RepoWise: A Conversational Framework for Open Source Repository Analysis with Retrieval-Augmented Generation and Structured Analytics," Software Engineering of Emerging Technologies, Communications in Computer and Information Science (CCIS), Springer, pp. 268-288, 2026. (**Best Paper Award**) [[LINK](#)]
9. **Nafiz Imtiaz Khan**, Tahasin Mahmud, and Muhammad Nazrul Islam, "COVID-19 and Black Fungus: Analysis of Public Perceptions through Machine Learning," Engineering Reports, 2022. (**IF = 2 and SJR Rank = Q2**) [[LINK](#)]
10. Zeerak Babar, **Nafiz Imtiaz Khan**, Muhammad Hassanain, Vladimir Filkov, "Open-Source LLMs for Technical Q&A: Lessons from StackExchange," International Conference on Software Engineering of Emerging Technologies (SEET), Springer, pp. 615-626, 2025. [[LINK](#)]
11. Lin Zhou, **Nafiz Imtiaz Khan**, Charlie Schweik, Vladimir Filkov, "Open-Source Software Sustainability: Defining and Measuring a Multifaceted Concept," (Under review). [[LINK](#)]
12. Amrin Rahman, Mahedi Kamal, Afrina Zinia, Amrin Akhter Pinky, Sabrina Afrin Toma, Abedur Rahman Khan, M Akhtaruzzaman, **Nafiz Imtiaz Khan**, Muhammad Nazrul Islam, "AI Integrated Entertaining and Monitoring System for Daily Care of Elders," Proceedings of the 3rd International Conference on Computing Advancements, 2024. [[LINK](#)]